

# Deva Anand

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## Education

### University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

**Coursework:** Advanced NLP, Advanced Machine Learning, Systems for Data Science, Optimization in CS, Performance Engineering

### Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

## Skills

### Languages

Python, TypeScript, JavaScript, SQL, C++, Java, Bash, HTML/CSS

### LLM & Agent Orchestration

LLM agents, multi-agent systems, tool use, function calling, agent orchestration, durable workflows, provider routing/fallback, prompt design, evaluation, OpenAI / Groq APIs, Ollama, MCP

### Full-Stack & Real-Time UIs

React, Electron, WebSockets, REST APIs, async services, data-intensive state management, FastAPI, Node.js, Express

### Data & Storage

MongoDB, PostgreSQL, Redis, SQLite, Apache Spark / PySpark, Databricks, sentence-transformers, RAG

### Cloud, Infra & DevOps

AWS, GCP, Azure, Docker, Cloudflare Workers / Durable Objects / D1, Git, GitHub Actions, CI/CD, observability / instrumentation

## Experience

### Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected and shipped a **data-intensive full-stack tool** (React, Node.js, MongoDB) used daily by the AI team to ingest, label, and surface high-volume training data through reviewer dashboards with complex state management; cut manual operations by **100+ hours/month** and lifted throughput **60%**.
- Owned multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**, integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines; hardened auth, rate limiting, and remediated **20+ VAPT findings** (contributed to **\$340K+** in revenue).

### Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Shipped **LLM-powered context-aware replies** into a customer-facing chat product, iterating on prompt design and response-quality heuristics from real usage; designed **20+ production RESTful APIs** in Node.js + Fastify over PostgreSQL with clear contracts and **JS→TS migration** across multiple repositories.

## Projects

### cf-ai-research-scout - Real-Time AI Agent with Durable Orchestration

github.com/Deva-1903/cf\_ai\_research\_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an **AI agent** end-to-end with a TypeScript frontend that fetches web sources, generates multi-step research digests, and **streams agent-generated insights to the client over WebSocket in real time**, surfacing live progress as the agent works.
- Designed durable workflow boundaries via WorkflowEntrypoint and Durable Objects so multi-step agent tasks survive process restarts and tool failures, with agent state and conversation history persisted in D1 without an external database.

### AgenticSearch - Multi-Stage Agent with Tool Integrations and Trust UI

github.com/Deva-1903/ciir\_agentic\_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic service with **tool integrations** (Brave Search, deterministic parsers, LLM fallback) and **agent orchestration** that converts open-ended queries into structured entity tables with cell-level provenance.
- Hardened the service with **150+ automated tests**, **provider routing/fallback across OpenAI and Groq**, pipeline instrumentation, and a reviewer-facing trust UI that surfaces agent decisions, source URLs, and ranking transparency as a product experience.

### Sonare - Full-Stack Real-Time Audio/Video Pipeline

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Took a **cross-platform desktop product** from zero to one under hackathon time pressure: React/Electron frontend, FastAPI backend, on-device pipeline integrating MediaPipe hand tracking, whisper.cpp ASR, and local TTS, with async APIs and queues keeping the UI responsive over high-volume video and audio streams.

## Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.